

Homework 5, 5.1

5.1) $V(t) = \sqrt{\frac{g m}{C_d}} \cdot \tanh\left(\sqrt{\frac{g C_d}{m}} \cdot t\right)$

$$0 = \sqrt{\frac{9.81(95)}{C_d}} \cdot \tanh\left(\sqrt{\frac{9.81 \cdot C_d}{95}} \cdot 9\right) - 46$$

0.2, 0.5

$0.2 - 0.35 = +$

$0.35 - 0.5 = -$

0.35, 0.5

$0.35 - 0.425 = -$

0.35, 0.425

$0.35 - 0.3875 = +$

$0.3875 - 0.425 = -$

0.40625

→

$$\frac{0.40625 - 0.3875}{0.40625}$$

· 100 =

4.6% error